



Precalculus A Study Guide

Credit by Exam for Credit Recovery or Acceleration

The exam you are interested in taking is designed to test your proficiency in the relevant subject matter. You should be thoroughly familiar with the subject matter before you attempt to take the exam. This CBE CR/A Study Guide can help you prepare for the exam by giving you an idea of what you need to review. You can check your familiarity level by reviewing the Texas Essential Knowledge and Skills (TEKS) for this course. (See below.) To refine your skills, you can refer to any of the state-adopted textbooks.

Texas Essential Knowledge and Skills (TEKS)

Every question that appears on this exam is derived from the knowledge and skills statements and student expectations within the Texas-mandated standards, the Texas Essential Knowledge and Skills (TEKS). You can view the TEKS for this exam online via the following link:

<http://ritter.tea.state.tx.us/rules/tac/chapter111/ch111c.html#111.42> Refer to section (c), Knowledge and skills, 1A–5N. Throughout this guide, you'll see TEKS references. These refer to the numbers listed under (c) Knowledge and skills; for example, 1A or 3B.

Materials Needed

You will need to bring a graphing calculator. The formula sheet at the end of this study guide will be provided during the exam. You can also bring this formula sheet with you to your exam. If you are taking a print exam, you must bring a #2 pencil to complete the exam. You will receive a computer-graded answer sheet when you arrive at the testing center. The proctor will provide scratch paper.

Exam Structure

You will be allowed **3 hours** to complete this exam. The Precalculus A exam consists of 73 multiple-choice questions worth a total of 100 points. The exam covers a wide variety of topics. To help you study, we have isolated 4 key topics and provided study tips and sample questions for each. You can expect several multiple-choice questions on each of the following topics:

Topic 1: Plan and Solve Real-world Problems

Topic 2: Families of Functions and Their Graphs

Topic 3: Trigonometry

Topic 4: Solving Problems Using Rational, Polynomial, Logarithmic, and Exponential Equations

Formula Chart

You will be provided with the formula chart at the end of this study guide. Familiarize yourself with these formulas and how to use them to solve problems based on the TEKS above.

Scholastic Honesty

When you arrive at the testing center you will be asked to carefully read the exam rules and sign a statement agreeing to take the exam in accordance with the rules. This is called the Examinee's Certification. The following is a copy of these rules:

Examinee's Certification

This certification must be signed *before* the exam is administered and then returned with the completed examination attached, or credit for the exam will not be given.

Scholastic dishonesty is a serious academic violation that will not be tolerated. Scholastic dishonesty encompasses, but is not limited to:

- copying from another student's work;
- using an unauthorized testing proctor or taking the exam at an unauthorized testing location;
- using materials not authorized by a testing proctor;
- possessing materials that are not authorized by a testing proctor, such as lessons, books, or notes;
- knowingly using or soliciting, in whole or Topic, the contents of an unadministered test;
- collaborating with or seeking aid from another student without authorization during the test;
- substituting for another person, or permitting another person to substitute for oneself, in taking a course test or completing any course-related assignment;
- using, buying, stealing, or transporting some or all of the contents of an unadministered test, test rubric, homework answer, or computer program.

Evidence of scholastic dishonesty will result in a grade of *F* on the examination and an *F* in the course (if applicable).

At the testing center, you will be asked to sign a statement that says you have read the above and agree to complete the examination with scholastic honesty.

General Study Tips

The bulleted lists and sample questions in this study guide can assist you in preparing for the exam. It is a fairly complete guide for studying, but does not cover every item on the test. Ultimately, you should use the TEKS to guide your exam preparation.

Additional Study Tips

The following information provides direction for your studies. For each part, you will find study tips and sample questions to give you a general idea of the types of questions you can expect to see on the exam.

Topic 1: Plan and Solve Real-world Problems

This is a broad topic that can relate to any type of question. Students should show the ability to create a plan for problem-solving, explain their reasoning, and solve problems arising in everyday life.

Study Tips for Topic 1:

This topic relates to TEKS 1A–1G. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- Solve word problems about the workplace and every day interactions in society
- Justify your solution to a problem
- Explain if a solution is or is not reasonable
- Create and use tables, graphs, number lines, Venn Diagrams and other representations to organize information

Sample Questions for Topic 1:

The following are sample questions. You can find the correct answers listed after the questions, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

1. The wind speed s (in miles per hour) near the center of a tornado is related to the distance d (in miles) the tornado travels. The speed can be found using a logarithmic model such that the speed is 65 more than 93 times the common log of the distance it has travelled. If a tornado with a wind speed of 200 miles per hour struck your city, how far did the tornado travel? Round your answer to the nearest mile.

A. 15 miles
B. 28 miles
C. 141 miles
D. 279 miles

Topic 2: Families of Functions and Their Graphs

This topic relates to your understanding of the basic parent functions and transformations of these functions. You are expected to know how to create a new function by composing two functions, to determine the inverse of a function, as well as to identify various properties of these functions and their graphs.

Study Tips for Topic 2:

This topic relates to TEKS 2A–2P. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- Use the composition of two functions to model and to solve real world problems
- Determine if order matters when composing functions (i.e. is function composition commutative? Why or why not?). You will be asked to demonstrate this using proper function notation
- Recognize and describe symmetry of even and odd functions
- Determine the inverse of a function when it exists
- Describe the domain of an inverse function and note whether the domain needs to be restricted
- Graph exponential, logarithmic, rational, polynomial, power, trigonometric, inverse trigonometric, and piecewise defined functions as well as transformations of each
- Describe a function in terms of transformations of the parent function
- Graph arcsine and arccosine functions and describe the restrictions on the domain and/or range
- Analyze and describe end behavior using of each parent function listed above using proper notation
- Analyze and describe behavior of rational functions as they approach asymptotes
- Determine various types of discontinuities and describe why certain types of discontinuities are not visible on a graphing calculator
- Use exponential, logarithmic, rational, polynomial, and power functions to model and solve real world problems
- Given a real-world problem, create a sinusoidal function to model and solve the problem
- Determine the value of the 6 trigonometric functions at special angles on the unit circle and use them to solve real world problems
- Familiarize yourself with the trigonometric ratios and unit circle given in the formula chart as well as how to use them to solve problems

Sample Questions for Topic 2:

The following are sample questions. You can find the correct answers listed after the questions, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the **BEST** responses to the following questions.

2. A post office in Rodgerstown, Wisconsin charges 34 cents to mail a package plus an additional cent for every ounce that the package weighs. Because they do not want to charge fractions of a cent, the register uses the function $\text{floor}(x)$ when weighing the packages. The function $\text{floor}(x)$ is defined as the largest integer that is not greater than x , where x is the weight of the package in ounces. What would be the exact cost for mailing a package weighing 55.7 ounces?
- A. 89 cents
B. 90 cents
C. 91 cents
D. 92 cents
3. Which of these represents the inverse function of $y = \sin(x + 70^\circ)$?
- C. $y = \sin^{-1}(x) - 70^\circ$
A. $y = \sin^{-1}(x + 70^\circ)$
B. $y = \sin^{-1}(x - 70^\circ)$
D. $y = 70^\circ - \sin^{-1}(x)$
4. If $f(x) = \frac{ax^2}{bx^2 + c}$, and as $x \rightarrow \pm\infty$, $f(x) \rightarrow 5$, which of the following must be true?
- A. $a = b$, and c can be any real number.
B. $a = 5b$, and c can be any real number.
C. $a = 5$, $b = 5$, and c can be any real number.
D. $a = 0$, $c = 5$, and b can be any real number.

Topic 3: Trigonometry

This topic relates to your understanding of the unit circle, angle measures in degrees and radians, and the use of trigonometric ratios and functions to solve problems.

Study Tips for Topic 3:

This topic relates to TEKS 4A–4E. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- Use the concept of periodic functions to recognize co-terminal angles (e.g. which angle on the unit circle is co-terminal with $1,560^\circ$?)
- Convert between radian and degree angle measures and describe the relationship between the two

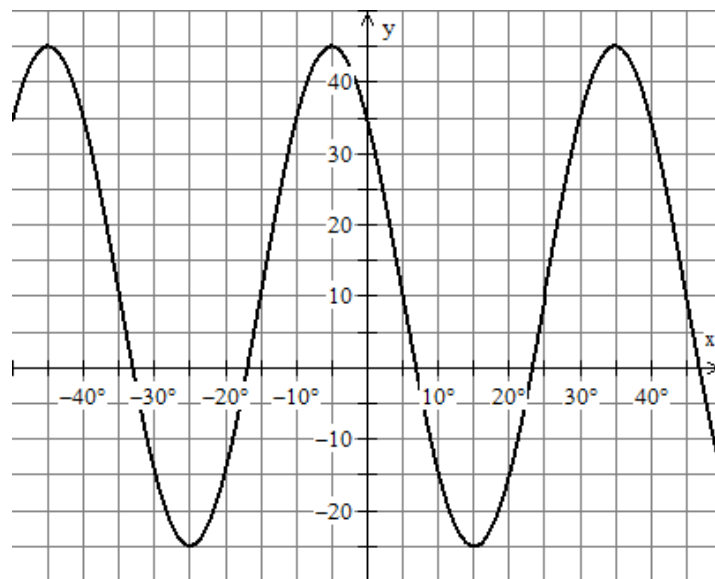
- Determine the measure of angles using standard position and directional bearing and use them to solve problems
- Determine the value of trig ratios of angles and use them to solve real world problems

Sample Questions for Topic 3:

The following are sample questions. You can find the correct answers listed after the questions, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the **BEST** responses to the following questions.

5. Which of the following statements is not correct, given the graph of the sinusoidal function, $y = f(x)$, below?



- A. $f(35^\circ) = 45$
- B. $f(75^\circ) = 45$
- C. $f(195^\circ) = 45$
- D. $f(225^\circ) = 45$

6. The hour hand of a standard clock moves with an angular velocity of 30° per hour. Which of these represents the angle of rotation for the hour hand between 1am and 4pm?
- A. $\frac{\pi}{2}$ radians
B. $\frac{5\pi}{2}$ radians
C. $\frac{2\pi}{3}$ radians
D. $\frac{7\pi}{3}$ radians
7. If $\sec(\theta) = 2$, then which of the following cannot be true?
- A. $\cos(\theta) = \frac{1}{2}$
B. $\cot(\theta) = \sqrt{3}$
C. $\sin(\theta) = \frac{\sqrt{3}}{2}$
D. $\csc(\theta) = \frac{2\sqrt{3}}{3}$

Topic 4: Solving Problems Using Rational, Polynomial, Logarithmic, and Exponential Equations

This topic relates to your understanding of using properties to simplify expressions and to solve equations. You will be expected to know properties and equation solving strategies for rational, polynomial, logarithmic, and exponential functions.

Study Tips for Topic 4:

This topic relates to TEKS 5G–5L. Familiarize yourself with those TEKS, and then be prepared to demonstrate knowledge of the following topics:

- Use properties of logarithms to evaluate, expand, and condense logarithmic expressions
- When given a word problem, generate and solve logarithmic and exponential equations
- Solve polynomial equations using multiple techniques
- Solve polynomial inequalities and check the solution intervals to verify the solution set(s)
- Solve rational inequalities in real world problems and write the solution set in interval notation

Sample Questions for Topic 4:

The following are sample questions. You can find the correct answers listed after the questions, but try answering the questions without looking at the answers first to check your comprehension.

DIRECTIONS: Select the BEST responses to the following questions.

8. Evaluate the following expression: $3\log_2 4 - \log_2 32$.
- A. 0
 - B. 1
 - C. 2
 - D. 11
9. A major employer in a city went bankrupt and had to close, causing many people to lose their jobs and move to new cities. This event started a decline in the population. If the initial population of the city was 80,000 when the decline began, and the population decreased at a rate of 3% per year, how many years will it take for the population to be cut in half? Round to the nearest whole year.
- A. 13 years
 - B. 20 years
 - C. 23 years
 - D. 52 years
10. Which of the following satisfy the inequality $\frac{4}{x+4} < \frac{1}{x-5}$?
- I. $x = 8$
 - II. $x = 4$
 - III. $x = -7$
- A. I only
 - B. III only
 - C. I and II only
 - D. II and III only

Answer Key

Item Number	Correct Answer	TEKS expectation
1	B	1A, 1B, 5H
2	A	2I, 2N
3	A	2E
4	B	2J, 2K
5	D	4A
6	B	4C, 4D
7	B	4E
8	B	5G
9	C	5I
10	B	5L

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PRE CALC A

Formulas

Even/Odd Formulas

$$\cos(-u) = \cos u$$

$$\sec(-u) = \sec u$$

$$\sin(-u) = -\sin u$$

$$\csc(-u) = -\csc u$$

$$\tan(-u) = -\tan u$$

$$\cot(-u) = -\cot u$$

Trigonometric Functions

$\sin \theta = \text{Opposite/Hypotenuse}$

$\cos \theta = \text{Adjacent/Hypotenuse}$

$\tan \theta = \text{Opposite/Adjacent}$

$\csc \theta = \text{Hypotenuse/Opposite}$

$\sec \theta = \text{Hypotenuse/Adjacent}$

$\cot \theta = \text{Adjacent/Opposite}$

Sector Area

$$A_s = \frac{\theta}{360} \cdot \pi r^2 \text{ where } \theta \text{ is measured in degrees or } A_s = \frac{\theta r^2}{2} \text{ where } \theta \text{ is measured in radians}$$

Polar Equations

Polar Coordinates

(r, θ)

$$r = \sqrt{a^2 + b^2}$$

$$\theta = \arctan\left(\frac{b}{a}\right)$$

Rectangular coordinates

(x, y)

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Circles

$$(x - h)^2 + (y - k)^2 = r^2$$

Unit Circle

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